THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Temperature and its Measurement** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which temperature is not possible on a clinical thermometer scale? 1

(a) 36 °C **(b)** 42 °C **(c)** 40 °C **(d)** 20 °C

2. The difference between heat and temperature, as at this level, is that temperature tells: 1

(a) how hot something is **(b)** its colour
(c) how much it weighs **(d)** its volume

3. Which statement is correct? 1

(a) a larger object always has higher temperature
(b) a bigger object is not always hotter
(c) cold objects have no heat
(d) temperature depends on size

4. A student wraps ice in a woollen cloth and another piece in a metal foil. Which melts first? 1

(a) the one in metal foil **(b)** the one in wool
(c) both equal **(d)** neither

5. Which of the following is the correct sequence of temperatures from lowest to highest? 1

(a) body, room, ice, boiling **(b)** ice, room, body, boiling water
(c) room, ice, body, boiling **(d)** boiling, body, room, ice

6. A pan of boiling water and a bathtub of warm water: which has higher temperature? 1

(a) equal **(b)** boiling water **(c)** cannot say **(d)** bathtub

7. Which of these helps in storing ice cream for long? 1

(a) open plate **(b)** aluminium plate **(c)** iron box **(d)** thermocol box

8. A cup of tea at 90 °C loses heat quickly at first and slowly later. The reason is that: 1

(a) the cup is heavy
(b) the temperature difference with the room reduces
(c) heat is absorbed
(d) the room is cold

9. A thermometer placed in direct sunlight shows a higher reading than in the shade because: 1

(a) the scale changes **(b)** the bulb absorbs sunlight
(c) mercury increases **(d)** the air is hotter

10. The bulb of a clinical thermometer is small so that it: 1

(a) looks small **(b)** costs less

(c) is cooler **(d)** quickly reaches the body temperature

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Describe why a cloth soaked in cold water on the forehead helps reduce fever. 2

12. Why is a thermometer's reading checked with fixed points? 2

13. Why can the temperature of boiling water not be raised above 100 °C in an open vessel on heating continuously? 2

14. Why is it necessary to dry the lab thermometer after dipping in a hot liquid before putting in cold? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cup of tea was cooled from 80 °C to 40 °C in 10 minutes. It cooled faster in the first 5 minutes than in the next. Explain. 3

16. Why do metals feel colder than wood at 15 °C but warmer than wood at 50 °C? 3

17. Why does a metal bottle sweat on the outside when filled with cold water? 3

OR

Explain, with reasons, why we should not place a mercury thermometer in boiling water directly without checking its range. 3

18. A hot piece of iron (100 °C) is dropped into 1 L of water at 30 °C. State what happens to the temperatures. 3

OR

Explain why a layer of snow can keep the ground warm. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A thermos flask has two glass walls with vacuum, silvered walls and a cork. Explain how it reduces heat loss by each part. 5

OR

A scientist wants a material to keep medicines cold for a day while travelling. Suggest the container design with reasons. 5

20. Analyse the data: time (min) 0, 5, 10, 15 ; temperature (°C) 80, 60, 48, 42 for a cooling cup. Describe the pattern and predict the trend. 5

OR

Explain how the sense of touch judges rate of heat flow rather than temperature, using iron and wood examples. 5

END OF QUESTION PAPER Total: 40 Marks